Session 6

Q. 1) Create a SQL query using GROUP BY to show the total number of orders placed by each user in an ‘orders’ table, displaying user_id and order_count.

Ans. create table orders

(order_id int(11) auto_increment primary key, user_id int(11));

insert into orders

(user_id)

values

(1),(1),(1),(1),(2),(2),(2),(3),(3),(4),(5),(5);

select user_id, count(*) as order_count from orders group by user_id;

Q. 2) Write a SQL query to display the names of restaurants from a ‘restaurants’ table that have received an average rating above 4.0, using GROUP BY and HAVING.

Ans. select name from restaurants

group by name

having avg(rating) > 4.0;

Q. 3) Suppose you have a ‘payments’ table with columns user_id, amount and payment_date. Write a subquery to find user_ids who have made a single payment above 2000 rs. in any transaction.

Ans. create table payments

(user_id int(11) auto_increment primary key, amount int(11), payment_date date);

insert into payments (amount, payment_date)

values

(2500, '2026-11-5'),

(3500, '2026-11-6'),

(1500, '2026-11-7'),

(500, '2026-11-8'),

(5500, '2026-12-9');

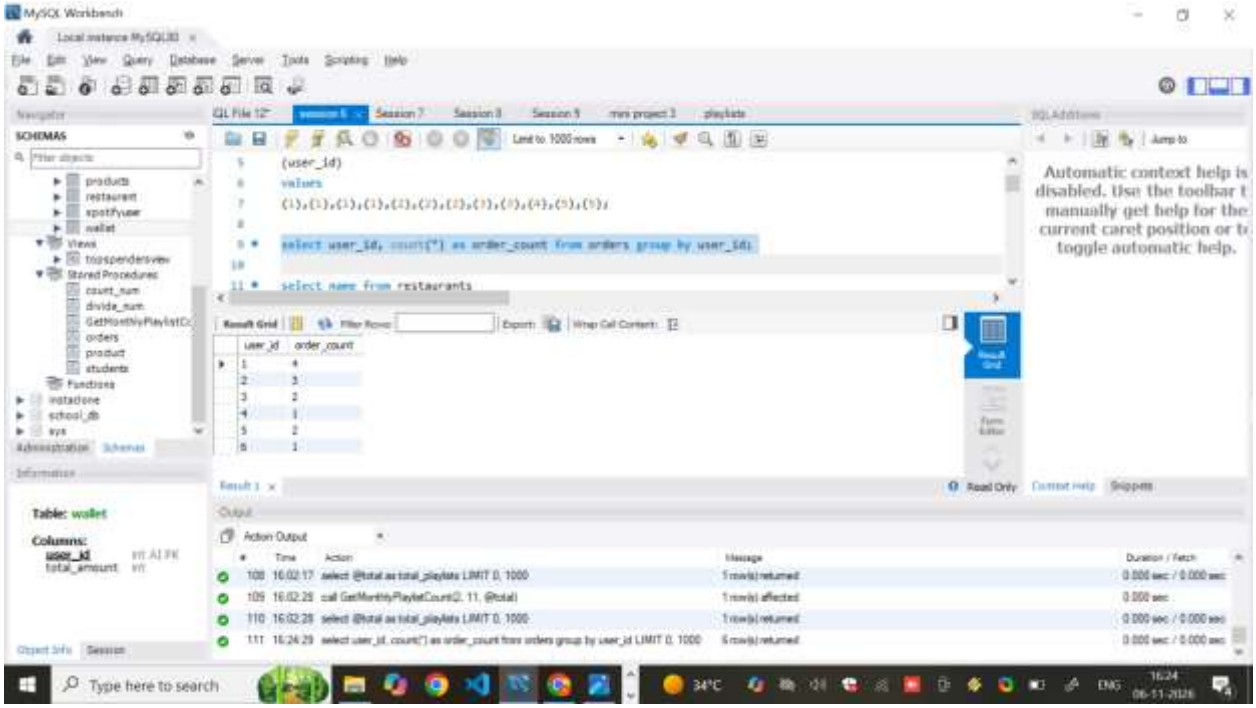
```
select user_id from orders
where user_id
in (select user_id from payments where amount > 2000);
```

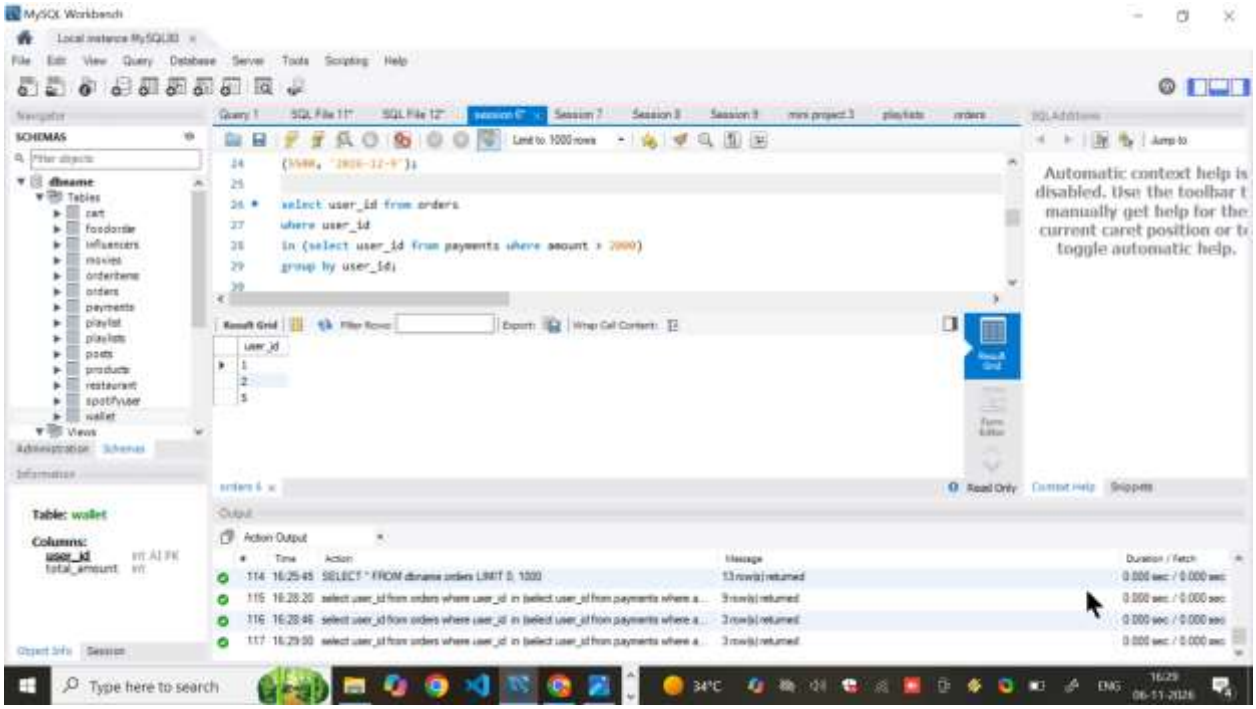
Q. 4) Using a nested subquery, display the names of movies from a ‘movies’ table that have a higher average rating than the overall average rating of all movies. Hint: Use a subquery to calculate the overall rating, then compare each movie’s average rating against it in the main query.

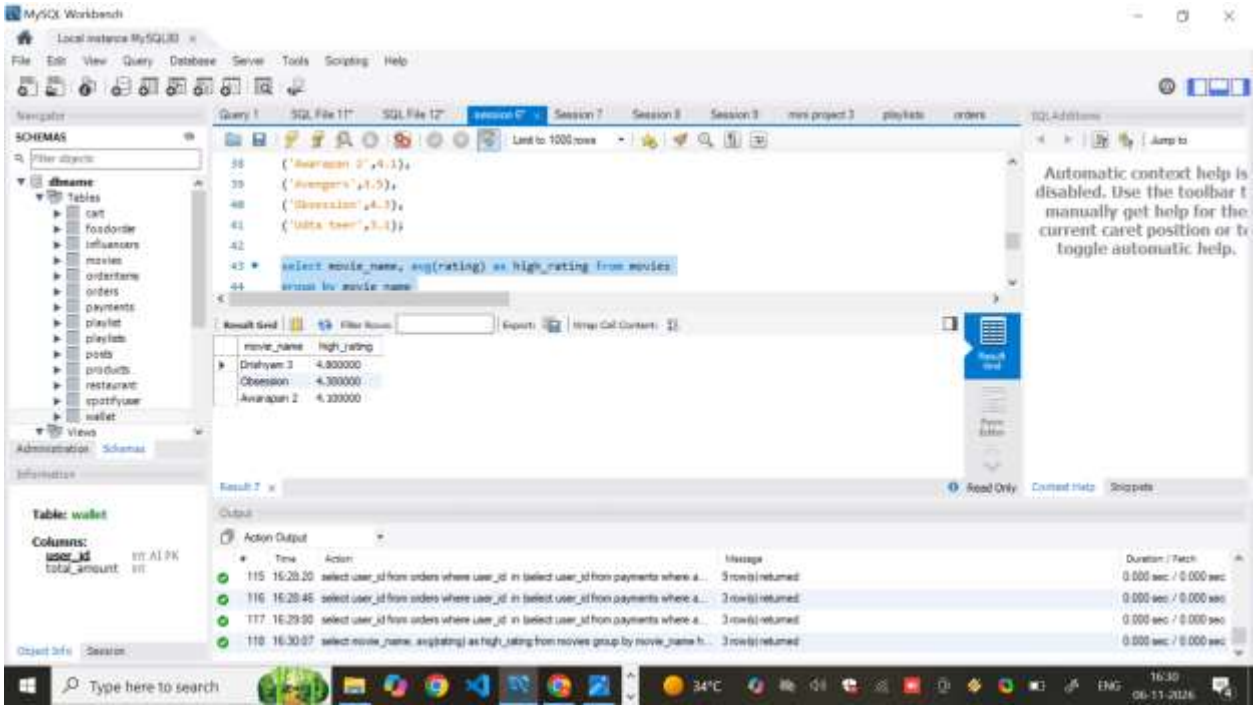
Ans.
```
create table movies
(movie_id int auto_increment primary key, movie_name varchar(100), rating decimal(3,2));

insert into movies
(movie_name, rating)
values
('Drishyam 3',4.8),
('Awarapan 2',4.1),
('Avengers',3.5),
('Obsession',4.3),
('Udta teer',3.1);

select movie_name, avg(rating) as high_rating from movies
group by movie_name
having avg(rating) > (select avg(rating) from movies)
order by high_rating desc;
```


MySQL Workbench
select user_id, count(*) as order_count from orders group by user_id;
select name from restaurants
user_id order_count
1 4
2 3
3 2
4 1
5 2
6 1
Table: wallet
Columns:
user_id
total_amount
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Action Output
108 16:02:17 select @total as total_playlists LIMIT 0, 1000 1 row(s) returned
109 16:02:28 call GetMonthlyPlaylistCount(2, 11, @total) 1 row(s) affected
110 16:02:28 select @total as total_playlists LIMIT 0, 1000 1 row(s) returned
111 16:24:29 select user_id, count(*) as order_count from orders group by user_id LIMIT 0, 1000 6 row(s) returned
Type here to search


MySQL Workbench
select user_id from orders
where user_id
in (select user_id from payments where amount > 2000)
group by user_id;
user_id
1
2
5
Table: wallet
Columns:
user_id
total_amount
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Action Output
114 16:25:45 SELECT * FROM dbname.orders LIMIT 0, 1000 13 row(s) returned
115 16:28:20 select user_id from orders where user_id in (select user_id from payments where a... 5 row(s) returned
116 16:28:46 select user_id from orders where user_id in (select user_id from payments where a... 3 row(s) returned
117 16:29:00 select user_id from orders where user_id in (select user_id from payments where a... 3 row(s) returned
Type here to search


MySQL Workbench
('Avarapan 2',4.1),
('Avengers',4.5),
('Obsession',4.3),
('Udta teer',3.1);
select movie_name, avg(rating) as high_rating from movies
group by movie_name
movie_name high_rating
Drishyam 3 4.800000
Obsession 4.300000
Avarapan 2 4.100000
Table: wallet
Columns:
user_id
total_amount
Automatic context help is disabled. Use the toolbar t manually get help for the current caret position or t toggle automatic help.
Type here to search
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